

Shiham Chowdhury

 Shiham |  Shiham Chowdhury |  shiham@umich.edu |  Portfolio

EDUCATION

University of Michigan

Bachelor of Science in Electrical Engineering

August 2022 - May 2026

Ann Arbor, MI

TECHNICAL SKILLS

- **Languages:** Java, Javascript, C++, Python, MATLAB
- **Developer Tools:** Visual Studio, Jupyter Notebook, LTSpice, React, Git
- **Hardware:** Soldering, Oscilloscope, PCB Design, Network Analyzer

PROJECTS

Environment Monitoring Device | *Arduino IDE, Amazon Web Services (AWS)*

[Link to Project](#)

- Developed an IoT-based environment monitor using ESP32 and DHT11 sensors to collect and analyze climate data.
- Integrated AWS IoT services to secure data transmission and enable both cloud-based analytics and remote monitoring.
- Averaged 98% accuracy on temperature and humidity readings over a two week testing period.

Word Occurrence Prediction Algorithm | *C++*

[Link to Project](#)

- Built a C++ algorithm that processes and learns language to predict the natural occurrence of a word.
- Utilized machine learning algorithms to analyze and categorize words into pair types for improved prediction accuracy.
- Employed recursive binary search trees to efficiently parse and manage learned data, facilitating quick computations of log-probability scores.

Euchre Game | *C++*

[Link to Project](#)

- Designed and implemented a fully functional Euchre game in C++ that supports up to 4 players.
- Integrated comprehensive game logic to enforce Euchre rules, handle gameplay flow, and ensure accurate game state management.
- Implemented strategic decision-making algorithms based on game state to simulate CPU opponents.

Temperature-Controlled Fan | *Arduino IDE, Arduino Uno*

[Link to Project](#)

- Designed and built an Arduino-based device to control a fan's operation based on temperature readings and proximity detection.
- Utilized a DHT11 sensor to monitor temperature, with data displayed on a 16x2 LCD screen for real-time feedback.
- Implemented logic to activate DC motor-powered fan when temperature is high enough and deactivate when temperature is low or when ultrasonic sensor detects an object getting too close.

EXPERIENCE

Orlans Junior Associate

June 2022 - present

- Optimized data workflows within the company's information database to ensure efficient processing.
- Performed comprehensive quality checks to verify documentation complied to industry standards.
- Ensured timely completion of tasks and deliverables, maintaining workflow deadlines and efficiency.